Comparative Study of Bayesian and Frequentist Shrinkage: Regression and Vector Autoregression Analysis

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Github Repository: Bayesian-Regression-With-Structutred-Priors

Group No. 5

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Abstract

This project investigates the performance of Frequentist and Bayesian shrinkage methods in high-dimensional linear regression and Vector Autoregression (VAR). While Maximum Likelihood Estimation (MLE) often fails or overfits when the number of predictors p exceeds the number of observations n, Maximum A Posteriori (MAP) estimation utilizes prior distributions to regularize parameters and induce sparsity. We benchmark classical shrinkage models—including Normal, Laplace (Lasso), and Bayesian Elastic Net—against purely Bayesian hierarchical priors like the Horseshoe and Spike-and-Slab. Our experiments span synthetic datasets, real-world regression benchmarks, and VAR forecasting for macroeconomic data.

1 Introduction

Modern statistical modeling often encounters the $p \gg n$ regime, where the design matrix $X^T X$ becomes singular, making Ordinary Least Squares (OLS) estimation impossible. Multicollinearity further compounds the issue, as regressors become strongly correlated, making it difficult to disentangle their individual effects.In VAR settings, the parameter count scales as $d^2 p$, leading to severe overparameterization.

Shrinkage addresses this by deliberately introducing bias to reduce variance, improving conditioning, and enhancing predictive robustness. While Frequentist approaches impose deterministic penalties on coefficients, Bayesian methods encode shrinkage through prior distributions, enabling posterior uncertainty quantification and adaptive behavior.

2 Methodology and Models

The project evaluates three primary shrinkage architectures:

- **Global Shrinkage (Ridge/Normal):** Imposes an ℓ_2 penalty or Gaussian prior to stabilize estimation under multicollinearity and dense signals.
- **Sparse Shrinkage (Lasso/Laplace):** Uses ℓ_1 penalties or Laplace priors to drive small coefficients to zero, performing variable selection.
- **Group-Aware Shrinkage (Bayesian Elastic Net):** Employs a hierarchical compromise between L_1 and L_2 penalties to provide a "grouping effect" for correlated predictors.
- **Adaptive Global–Local (Horseshoe):** Utilizes heavy-tailed priors to strongly shrink noise while minimally penalizing large signals.
- **Exact Sparsity (Spike-and-Slab):** Employs discrete mixture priors for probabilistic variable inclusion.

3 Evaluation Criteria

Performance is evaluated along four dimensions:

1. **Estimation Accuracy:** Parameter RMSE and recovery of zero vs non-zero coefficients.

2. **Sparsity Recovery:** Support identification and false discovery rates.

3. **Predictive Performance:** Out-of-sample RMSE and forecast stability.

4. **Uncertainty Calibration:** Empirical coverage and interval length of credible intervals.

4 Regression Experiments

4.1 Synthetic Data Analysis

Ten synthetic datasets were generated to isolate effects such as heavy-tailed noise, heteroscedasticity, and block correlation.

Table 1: Synthetic Dataset Characteristics

Dataset	Characteristics
Sparse High-Dim	$p = 500, n = 100$, 98% zeros
Correlated	Block correlation structure
Dense Signal	Many small coefficients
Heavy Tailed	Noise follows t_3 distribution
Struct. Sparsity	Hierarchical/Group patterns

4.2 Real-World Benchmarks

EDA on clinical and genomic datasets revealed distinct geometric challenges:

- **Diabetes:** Low-dim baseline with moderate multicollinearity ($\kappa \approx 470$).
- **Riboflavin:** Extreme $p \gg n$ genomic setting ($n = 71, p = 4088$) with rank deficiency.
- **Arcene:** Ultra-high-dim classification ($p = 10,000$) with strong scale heterogeneity.

5 Vector Autoregression (VAR)

5.1 Synthetic Data Analysis

A synthetic multivariate time series was generated using a VAR(1) data-generating process (DGP). The transition matrix was randomly constructed with controlled sparsity and then stationarized; model fitting used a lag-stacked design matrix for a VAR(p_{fit}), allowing intentional overfitting when $p_{\text{fit}} > 1$.

5.2 Real-World Benchmarks

- The levels of employment (e), labor productivity (prod), and real wages (rw) exhibit clear upward trends, while unemployment (U) shows cyclical movement, motivating first-differencing for stationarity.

Table 2: VAR Synthetic Data (Scenario 1)

Component	Specification / Output
DGP (VAR(1))	$Y_t = AY_{t-1} + \varepsilon_t$
Noise	$\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, $\Sigma = 0.1\, I_d$
Matrix A	Randomly generated & stationarized
Design matrix (VAR(p_{fit}))	$X_t = [Y_{t-1}^\top,\ Y_{t-2}^\top,\ \ldots,\ Y_{t-p_{\text{fit}}}^\top]$
Training shapes	$X \in \mathbb{R}^{(T_{\text{train}} - p_{\text{fit}}) \times (d\, p_{\text{fit}})}$
True coefficients under overfitting	$B_{\text{true}} = [\, A \;\; 0 \;\; \cdots \;\; 0 \,]$
Scenario 1 setup	$d = 3$, $p_{\text{fit}} = 4$ (overfit)
Y shape	(200, 3)
A_{true} shape	(3, 3)
X_{train} shape	(196, 12)
Y_{train} shape	(196, 3)
B_{true} shape	(3, 12)
Sparsity in A_{true}	6/9 nonzero entries
Stability check	$\max \lvert\lambda(A_{\text{true}})\rvert = 0.3724$

- The differenced series fluctuate around zero with reduced trend, suggesting the main non-stationarity is removed after taking first differences.
- ACF/PACF plots for Δe, Δprod, Δrw, and ΔU show mostly short-run dependence (dominant early-lag spikes) with correlations dying out quickly.
- Cross-correlation of differenced variables indicates a strong negative association between Δe and ΔU (-0.856), and a moderate positive link between Δrw and ΔU (0.413).
- Other contemporaneous correlations are relatively weak (e.g., Δe–Δprod $= 0.189$, Δprod–$\Delta U = -0.220$), supporting a multivariate VAR analysis on differenced data.

6 Conclusion

Ridge and Gaussian priors excel in dense or highly correlated regimes. However, the Horseshoe prior is the most reliable architecture for ultra-high-dimensional sparse recovery. This study clarifies that the choice of shrinkage must be aligned with the specific high-dimensional geometry and noise characteristics of the data.

References

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[8] George, E. I., & McCulloch, R. E. (1993). Variable Selection Via Gibbs Sampling. *Journal of the American Statistical Association*, 88(423), 881-889. https://www.jstor.org/stable/2290777

[9] **Project Documentation:**

- Synthetic Data Analysis.pdf
- VAR Simulation Pipeline Summary.pdf
- High Dimensional EDA Report.pdf
- Canada VAR Pipeline Summary.pdf